

Quiz 3 Section 2

Student Name: _____

Student ID: _____

A factory produces small electronic components, and each component has a probability $p = 0.04$ of being defective.

Tasks (5 points each)

1. Write down the probability distribution for the number X of defective components in n independent trials, where p is the probability of defect in a single trial (give the probability mass function).

2. Find the probability that exactly 3 components are defective in a batch of 20 using the distribution given in Task 1.

3. Find the probability that at most 2 components are defective in a batch of 20 using the distribution given in Task 1.

4. Using the Poisson approximation, estimate the probability that exactly 4 components are defective when $n = 200$ and $p = 0.04$.